

Questions by topic: Soils: fertility, erosion & management - Set 1 (75 marks)

Answer all questions in the spaces provided.

0 1

Table 1 shows the textural composition of four soil samples.

Table 1

Soil sample	Sand / %	Silt / %	Clay / %
A	85	10	5
B	40	40	20
C	20	25	55
D	10	80	10

0 1

. 1 Identify the sample that is a loam and explain why loams are ideal for cultivating most crops.

[3 marks]

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. 2 Explain why sample A is likely to have the highest permeability but the lowest nutrient retention.

[3 marks]

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. 3 Compare the thermal capacity and root penetration of soils A and C.

[2 marks]

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0 1 . 4

Describe how the texture of a soil sample could be determined by sedimentation.

[2 marks]

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0 2 . 1

Explain the roles of **three** named groups of soil organisms in maintaining soil fertility.

[6 marks]

0 2 . 2

Explain why a soil pH between 5.5 and 7.0 is best for fertility.

[2 marks]

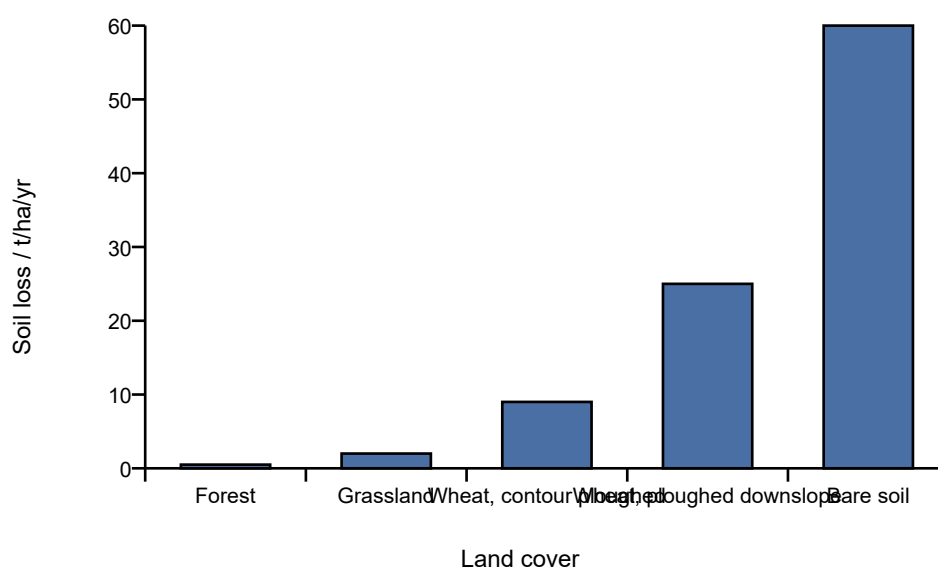
0 2 . 3

Explain how soil compaction by heavy machinery reduces fertility.

[2 marks]**10**

0 3

Figure 1 shows the annual soil loss measured on experimental plots with different vegetation cover on a 10% slope in a region with heavy seasonal rain.

Figure 1

0 3 . 1

Describe the pattern shown in Figure 1.

[2 marks]

03 . 2

Explain **four** ways in which vegetation reduces the rate of soil erosion.

[4 marks]

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03 . 3

Explain why ploughing up and down a slope causes more erosion than contour ploughing.

[2 marks]

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03 . 4

Explain why the region's climate makes the soil vulnerable to **both** wind and water erosion.

[2 marks]

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0 4 . 1

Explain how each of the following human activities increases soil erosion.

[4 marks]

Overgrazing

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Reducing soil biota

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0 4 . 2

Explain why landslides become more likely on steep slopes some years after deforestation.

[2 marks]

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0 4 . 3

Describe **four** environmental impacts of accelerated soil erosion away from the field where it occurs.

[4 marks]

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The Universal Soil Loss Equation (USLE) is $A = R \times K \times LS \times C \times P$. Table 1 gives factor values for a farm.

Table 1

Factor	Description	Value
R	Rainfall erosivity for the farm	110
K	Soil erodibility: loam	0.67
LS	Slope length and gradient: 120 m, 6%	1.35
C (crop type x tillage)	Beans (0.50) with spring ploughing (0.90); or wheat (0.35) with mulch tillage (0.60)	see question
P	Ploughing up and down slope 1.0; contour ploughing 0.50; strip cropping with contour ploughing 0.25	see question

Table 2

Soil erosion class	Rate of soil loss / $\text{t ha}^{-1} \text{yr}^{-1}$
Low	under 5
Moderate	5-10
High	10-20
Very high	20-30
Severe	over 30

0 5**1**

Calculate the annual soil loss for beans with spring ploughing, ploughed up and down the slope. Give your answer to one decimal place.

[2 marks]**Answer**

t/ha/yr

0 5**2**

Calculate the annual soil loss for wheat with mulch tillage and strip cropping with contour ploughing.

[2 marks]

Answer

t/ha/yr

05 . 3

The natural rate of soil formation on the farm is $1 \text{ t ha}^{-1} \text{ yr}^{-1}$. Use your answers and Table 2 to evaluate the sustainability of the two farming systems.

[3 marks]

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05 . 4

Explain why the USLE gives only an estimate of soil loss.

[2 marks]

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05 . 5

Suggest which single factor in the USLE a farmer can change most easily.

[1 mark]

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Complete Table 1 to explain how each method reduces soil erosion.

Do not write out

the box

Method	How it reduces erosion	
Terracing		
Tied ridging		
Windbreaks / hedgerows		
Mulching		
Strip cropping		

Explain why long-term crops such as tea, coffee or orchards reduce erosion compared with annual arable crops.

[2 marks]

Describe **three** ways livestock can be managed to reduce soil erosion.

[3 marks]

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07

A student determined the water and organic matter content of a soil sample. Table 1 shows the results.

Table 1

Measurement	Mass / g
Empty crucible	25.0
Crucible + fresh soil	65.0
Crucible + soil after drying at 105 °C to constant mass	55.0
Crucible + soil after heating in a furnace at 550 °C	52.6

07

1

Calculate the percentage water content of the fresh soil.

[2 marks]

Answer

%

07

2

Calculate the percentage organic matter content of the dry soil.

[2 marks]

Answer %

0 7 . 3 Explain why the soil is dried to constant mass before the organic matter is burnt off.

[2 marks]

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0 7 . 4 Explain how the organic matter content affects the fertility and erosion risk of this soil.

[3 marks]

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0 7 . 5 Suggest **one** reason why two students measuring the same field might get different organic matter results.

[1 mark]

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0 8 . 1

Which soil property is most directly responsible for capillary rise of water from below? Tick one box.

[1 mark]

- A High sand content
- B High clay content
- C Crumb peds
- D High pH

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0 8 . 2

Distinguish between hygroscopic, capillary and gravitational water in soil and state which are available to plants.

[3 marks]

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0 8 . 3

Explain why crumb peds produce more fertile soil than platy peds.

[1 mark]

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END OF QUESTIONS